COMPLETE NUMERICAL WORKBOOK & ALGORITHM TRACE MANUAL

WORKBOOK PURPOSE & SYLLABUS ALIGNMENT

This workbook provides a comprehensive, step-by-step numerical and trace-based preparation manual for **Unit 2: Solving Problems by Searching**. Designed for B.Tech Computer Engineering (SVKM's NMIMS MPSTME & University Pattern), it covers every numerical category prescribed by the official course policy, including **Uninformed Search (BFS, DFS, UCS, DLS, IDS)**, **Informed Search (GBFS, A*, 8-Puzzle)**, **Local Optimization (Hill Climbing, Genetic Algorithm)**, and **Adversarial Game Trees (Minimax, Alpha-Beta Pruning)**.

WORKBOOK STRUCTURE & TABLE OF CONTENTS

Section	Topic Domain	Core Focus & Numerical Types
Section 1	Formula Sheet & Decision Guide	Complexity formulas, admissibility proofs, algorithm selection tree
Section 2	Uninformed Search Numericals	BFS, DFS, UCS, DLS, IDS traces with OPEN/CLOSED lists & costs
Section 3	Informed Search Numericals	GBFS, A* Search, 8-Puzzle (Misplaced tiles & Manhattan distance)
Section 4	Local Search & Optimization	Hill Climbing landscape pathologies & Worked Genetic Algorithm generation
Section 5	Adversarial Search Numericals	Minimax utility backing-up & Alpha-Beta pruning ($\alpha \geq \beta$ cutoffs)
Section 6	Exam Pitfalls & Common Mistakes	Trap avoidance receptacle & step-by-step solution verification rules
Section 7	Numerical Readiness Checklist	Practice problem priority list & 100% syllabus coverage audit

1. GENERAL NUMERICAL FORMULA SHEET & ALGORITHM GUIDE



Figure 1: Formal 5-Component Problem Formulation Framework

MASTER MATHEMATICAL & COMPLEXITY FORMULA SHEET

1. Uninformed Search Complexities:

- **Breadth-First Search (BFS):** Time = $O(b^{d+1})$, Space = $O(b^{d+1})$ | Complete & Optimal for unit step costs.
- **Depth-First Search (DFS):** Time = $O(b^{m})$, Space = $O(bm)$ | Incomplete in infinite spaces, non-optimal.
- **Uniform-Cost Search (UCS):** Time = $O(b^{1 + \lfloor C^* / \varepsilon \rfloor})$, Space = $O(b^{1 + \lfloor C^* / \varepsilon \rfloor})$ | Complete & Optimal for step costs $\geq \varepsilon > 0$.
- **Iterative Deepening Search (IDS):** Time = $O(b^{d})$, Space = $O(bd)$ | Combines linear space of DFS with optimality of BFS.

2. Informed Search Evaluation Functions:

- **Greedy Best-First Search (GBFS):** $f(n) = h(n)$ (pure heuristic drive, non-optimal).
- **A* Search:** $f(n) = g(n) + h(n)$ (path cost so far + estimated cost to goal).
- **Admissibility Condition:** $0 \leq h(n) \leq h^*(n)$ (heuristic never overestimates actual cost).
- **Consistency (Monotonicity) Condition:** $h(n) \leq c(n, a, n') + h(n')$ (triangle inequality along edges).

3. Genetic Algorithm & Game Tree Formulas:

- **GA Selection Probability:** $p_i = f_i / \Sigma f$ | **Expected Count:** $e_i = p_i \times N$.
- **Alpha-Beta Cutoff Condition:** Prune remaining subtrees at MAX node if $\alpha \geq \beta$; prune at MIN node if $\alpha \geq \beta$.

Problem Characteristic	Recommended Search Algorithm	Algorithmic Justification
All step costs equal (unit cost = 1)	**Breadth-First Search (BFS)**	Guarantees shallowest goal path with minimal evaluation overhead.
Varying non-negative step costs	**Uniform-Cost Search (UCS)**	Expands nodes in order of cumulative path cost $g(n)$.
Large search space & strict memory limit	**Iterative Deepening Search (IDS)**	Requires only linear space $O(bd)$ while guaranteeing optimality.
Heuristic $h(n)$ available & shortest path needed	**A* Search**	Combines $g(n)$ and $h(n)$; optimal when $h(n)$ is admissible/consistent.
2-Player zero-sum perfect information game	**Alpha-Beta Pruning**	Yields same optimal decision as Minimax but cuts time to $O(b^{m/2})$.

2. UNINFORMED SEARCH NUMERICALS & ALGORITHM TRACES

HOW TO APPROACH UNINFORMED SEARCH NUMERICALS

Inputs: State graph/tree, initial state s_0, goal state G, edge costs $c(s, a, s')$.
Data Structures: BFS = FIFO Queue | DFS = LIFO Stack | UCS = Priority Queue ordered by $g(n)$.
Goal-Test Location (CRITICAL EXAM TRAP!):

- **BFS:** Applied when node is *generated* (inserted into frontier).
- **DFS & UCS:** Applied when node is *selected for expansion* (popped from frontier).

Tie-Breaking Convention: Alphabetical order unless specified otherwise.
Solution Verification: Confirm path cost matches sum of edge weights and no explored nodes are re-expanded.

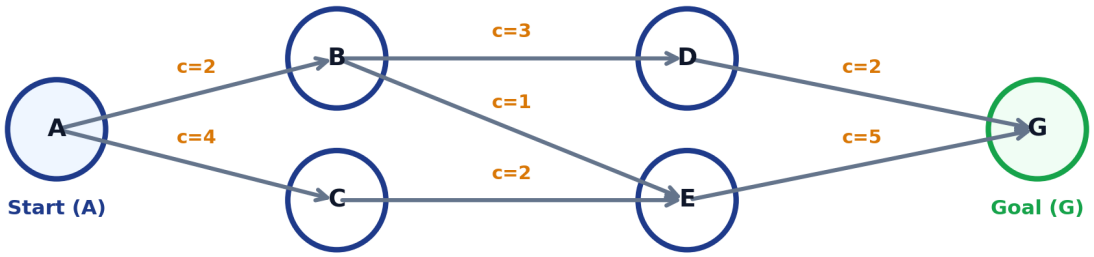


Figure 2: Benchmark Weighted Search Graph for BFS, DFS, and UCS Traces

Solved Numerical 2.1 (Hard — Uniform-Cost Search Trace) [10 Marks]

Problem Statement: Apply Uniform-Cost Search (UCS) on the weighted directed graph in Figure 2 to find the optimal path and path cost from Start Node **A** to Goal Node **G**. Show the complete step-by-step execution log including Priority Queue (Frontier) states, Explored Set (CLOSED), parent links, node expanded at each step, and final path cost.

Step	Node Expanded	Priority Queue (Frontier / OPEN) Format: Node(cost, parent)	Explored Set (CLOSED)
0	— (Start)	[A(g=0, parent=None)]	{ }
1	**A (g=0)**	[B(g=2, par=A), C(g=4, par=A)]	{ A }
2	**B (g=2)**	[E(g=3, par=B), C(g=4, par=A), D(g=5, par=B)]	{ A, B }
3	**E (g=3)**	[C(g=4, par=A), D(g=5, par=B), G(g=8, par=E)]	{ A, B, E }
4	**C (g=4)**	[D(g=5, par=B), G(g=8, par=E)] *(Note: E already in CLOSED, ignored)*	{ A, B, E, C }
5	**D (g=5)**	[G(g=7, par=D), G(g=8, par=E)] *(D->G cost=5+2=7 < existing 8; updated!)*	{ A, B, E, C, D }
6	**G (g=7)**	**GOAL REACHED & POPPED!**	{ A, B, E, C, D, G }

UCS SOLUTION SUMMARY & VERIFICATION

- **Expansion Sequence:** A → B → E → C → D → G.
- **Optimal Path:** A → B → D → G (Backtracked via parent links: G ← D ← B ← A).
- **Optimal Path Cost:** *g(G)* = 2 (A→B) + 3 (B→D) + 2 (D→G) = **7**.
- **Key Exam Insight:** Path A→B→E→G has cost 2+1+5=8. UCS successfully discovered the cheaper path A→B→D→G (cost 7) because goal test is applied when node G is *selected for expansion*!

3. INFORMED SEARCH NUMERICALS (GBFS, A*, 8-PUZZLE)

HOW TO APPROACH INFORMED SEARCH NUMERICALS

Evaluation Function:

- **Greedy Best-First Search (GBFS):** $f(n) = h(n)$ (Sorts OPEN purely by heuristic).
- **A* Search:** $f(n) = g(n) + h(n)$ (Sorts OPEN by total estimated path cost).

Admissibility Verification Step: Check that $h(n) \leq h^*(n)$ for every node.

8-Puzzle Heuristics:

- **Misplaced Tiles (h1):** Count of tiles in wrong positions (excluding blank).
- **Manhattan Distance (h2):** $\Sigma (|x_1 - x_2| + |y_1 - y_2|)$ for all non-blank tiles.

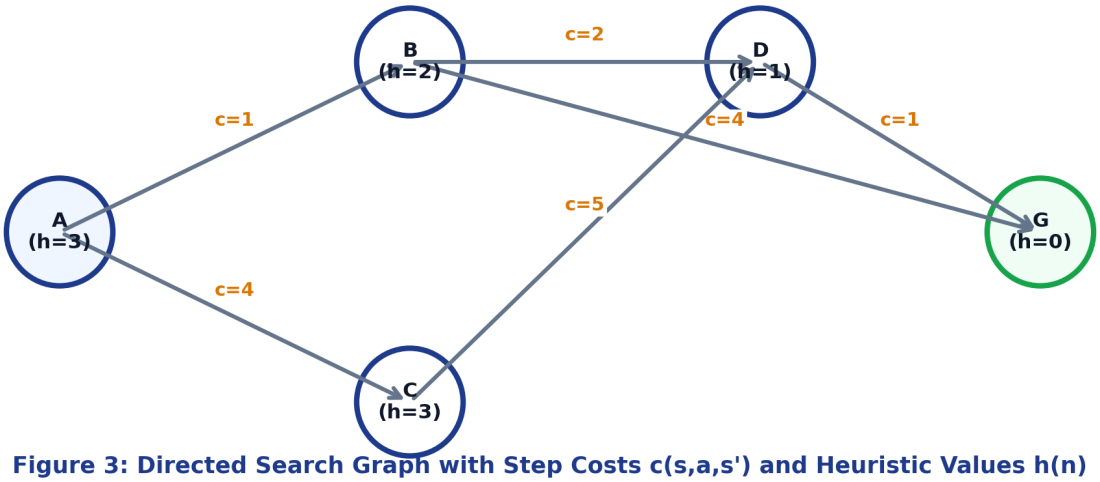


Figure 3: Directed Search Graph with Step Costs c(s,a,s') and Heuristic Values h(n)

Solved Numerical 3.1 (Actual PYQ — GBFS vs. A* Graph Search) [10 Marks]

Source: Compiled NMIMS B.Tech CE Final Examination, Batch 2024-25, Q2.c [10 Marks]

Problem Statement: Apply Greedy Best-First Search (GBFS) and A* Search on the directed graph in Figure 3 to find the path and path cost from Start Node **A** to Goal Node **G**. Verify heuristic admissibility.

Step	Expanded	GBFS Frontier OPEN [f=h]	A* Frontier OPEN [f=g+h]	CLOSED Set
0	—	[A(h=3)]	[A(g=0,h=3,f=3)]	{ }
1	**A**	[B(h=2), C(h=3)]	[B(g=1,h=2,f=3), C(g=4,h=3,f=7)]	{ A }
2	**B**	[D(h=1), C(h=3), G(h=0)] → **Pops G!**	[D(g=3,h=1,f=4), G(g=5,h=0,f=5), C(g=4,h=3,f=7)]	{ A, B }
3	**D (A*)**	— (GBFS Done)	[G(g=4,h=0,f=4), G(g=5,h=0,f=5), C(g=4,h=3,f=7)]	{ A, B, D }
4	**G (A*)**	— (GBFS Done)	**GOAL REACHED (A*)!**	{ A, B, D, G }

GBFS VS A* COMPARISON RESULT

• **GBFS Solution:** Path **A** → **B** → **G** | Path Cost = 1 + 4 = **5** (Suboptimal!).
• **A* Solution:** Path **A** → **B** → **D** → **G** | Optimal Path Cost = 1 + 2 + 1 = **4**.
• **Admissibility Proof:** Actual true cost to goal *h*(n)*: *h*(A)=4, h*(B)=3, h*(C)=6, h*(D)=1, h*(G)=0*.
Comparing given *h(n)*: *h(A)=3 ≤ 4, h(B)=2 ≤ 3, h(C)=3 ≤ 6, h(D)=1 ≤ 1, h(G)=0 = 0*. Since *h(n) ≤ h*(n)* for all nodes, *h(n)* is **strictly admissible**, guaranteeing A* optimality!

Solved Numerical 3.2 (Actual PYQ — 8-Puzzle A* Search Tree Trace) [10 Marks]

Source: Compiled NMIMS B.Tech CE Final Examination, Academic Year 2023-24, Q4.a [10 Marks]
Problem Statement: Trace A* Search to solve the 8-puzzle from Initial State **[2 8 3 / 1 6 4 / 7 _ 5]** to Goal State **[1 2 3 / 8 _ 4 / 7 6 5]** using Misplaced Tiles heuristic *h(n)*. Show all intermediate grids, *g(n)*, *h(n)*, *f(n)*.

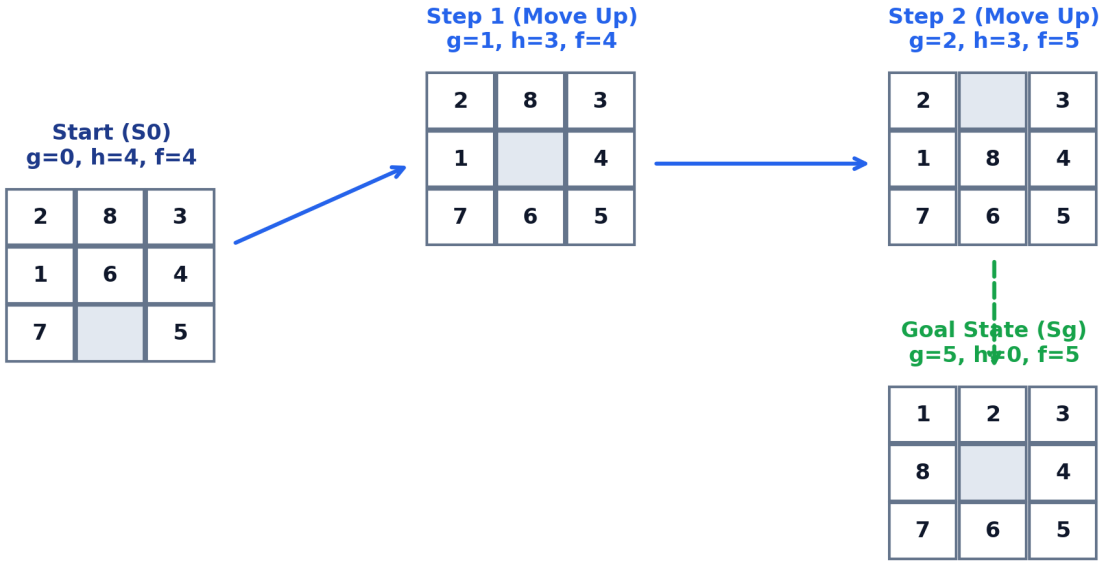


Figure 4: 8-Puzzle A* Search Tree Expansion with Grid States and f(n)=g(n)+h(n)

8-PUZZLE A* SEARCH SOLUTION TRACE

Step-by-Step 5-Move Optimal Path Trace:
• **Step 0 (Start S0):** 2 8 3 / 1 6 4 / 7 _ 5 | *g=0, h=4* {2,8,1,6 misplaced}, *f=4*.
• **Step 1 (Move Up S1):** 2 8 3 / 1 _ 4 / 7 6 5 | *g=1, h=3* {2,8,1 misplaced}, *f=4*. [SELECTED]
• **Step 2 (Move Up S2):** 2 _ 3 / 1 8 4 / 7 6 5 | *g=2, h=3* {2,1,8 misplaced}, *f=5*. [SELECTED]
• **Step 3 (Move Left S3):** _ 2 3 / 1 8 4 / 7 6 5 | *g=3, h=2* {1,8 misplaced}, *f=5*. [SELECTED]
• **Step 4 (Move Down S4):** 1 2 3 / _ 8 4 / 7 6 5 | *g=4, h=1* {8 misplaced}, *f=5*. [SELECTED]
• **Step 5 (Move Right Sg):** 1 2 3 / 8 _ 4 / 7 6 5 | *g=5, h=0* {All correct!}, *f=5*. **GOAL REACHED!**
Optimal Solution Cost: 5 moves.

4. LOCAL SEARCH & OPTIMIZATION (HILL CLIMBING & GENETIC ALGORITHM)

Figure 5: State-Space Landscape with Global Maximum, Local Maximum, and Plateau

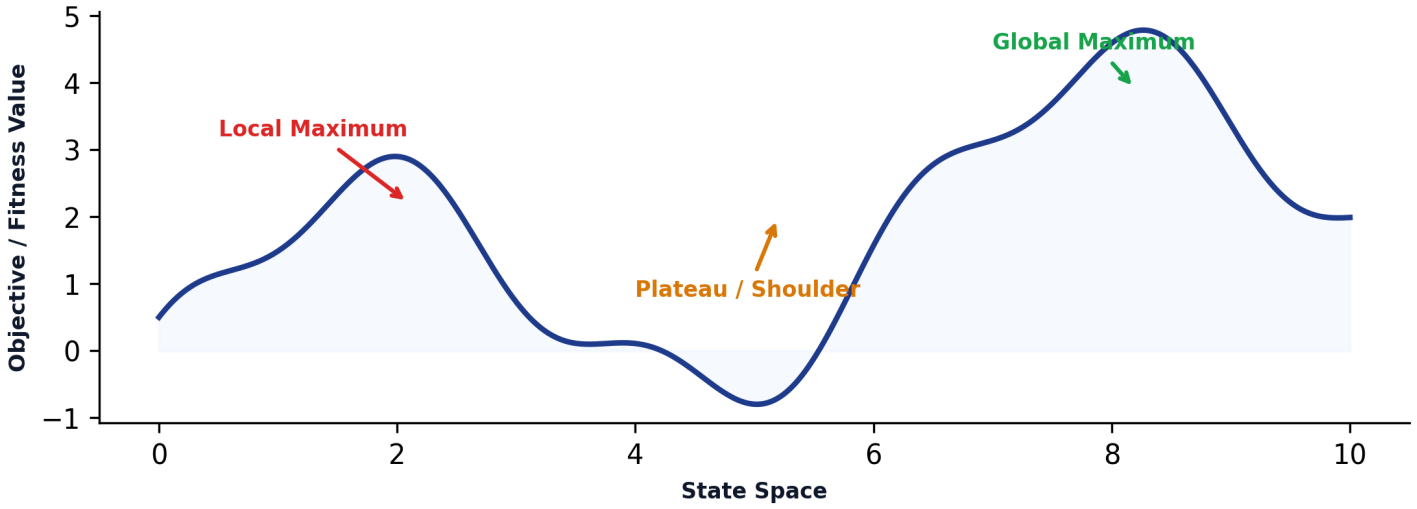
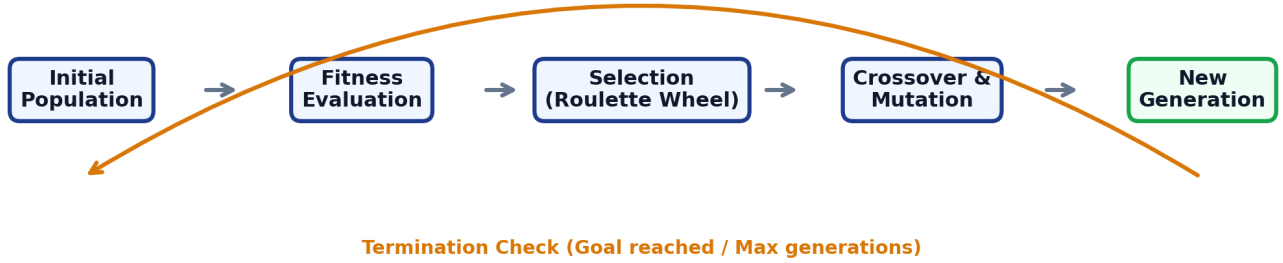


Figure 6: Genetic Algorithm Evolutionary Optimization Cycle



Solved Numerical 4.1 (Worked Genetic Algorithm Generation) [10 Marks]

Problem Statement: Given an initial population of 4 binary chromosomes representing 5-bit integers $x \in [0, 31]$: **S1 = 01100 (x=12)**, **S2 = 11001 (x=25)**, **S3 = 00011 (x=3)**, **S4 = 10011 (x=19)**. Fitness function is **$f(x) = x^2$**. Perform one complete GA generation: calculate fitness, selection probabilities, roulette wheel selection, single-point crossover at position 3, mutation at bit 4 for string 1, and show new generation average fitness improvement.

String No.	Initial Chromosome	Integer x	Fitness $f(x)=x^2$	Selection Prob $p_i = f_i / \Sigma f$	Expected Count e_i
1	01100	12	144	144 / 1139 = **12.6%**	0.51
2	11001	25	625	625 / 1139 = **54.9%**	2.20
3	00011	3	9	9 / 1139 = **0.8%**	0.03
4	10011	19	361	361 / 1139 = **31.7%**	1.27
Total / Avg	—	—	**Σf = 1139 | Avg = 284.8**	100.0%	4.00

GENETIC ALGORITHM GENERATION RESULT

Step-by-Step Evolutionary Reproduction Log:

• **Selection (Mating Pool):** Based on expected counts, String 2 receives 2 copies, String 4 receives 1 copy, String 1 receives 1 copy. (String 3 eliminated!). Mating Pool = {S2, S2, S4, S1}.

• **Crossover Pair 1 (S2=110|01 & S4=100|11 at pos 3):**

- Offspring 1' = 11011 (x=27, f=729) | Offspring 2' = 10001 (x=17, f=289).

• **Crossover Pair 2 (S2=110|01 & S1=011|00 at pos 3):**

- Offspring 3' = 11000 (x=24, f=576) | Offspring 4' = 01101 (x=13, f=169).

• **Mutation (Bit 4 flipped in Offspring 4'):** 01101 → 01111 (x=15, f=225).

• **New Generation Fitness Total:** $\Sigma f'$ = 729 + 289 + 576 + 225 = **1819** | **New Avg Fitness = 454.8**.

• **Performance Gain:** Average fitness increased from **284.8 to 454.8 (+59.7%)** in a single generation!

5. ADVERSARIAL SEARCH NUMERICALS (MINIMAX & ALPHA-BETA)

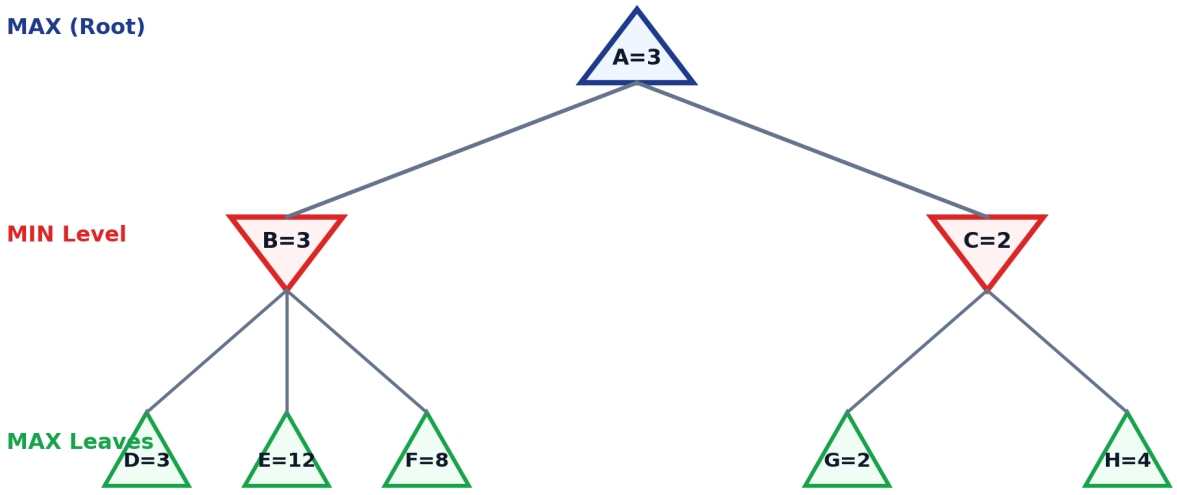


Figure 7: Solved Minimax Game Tree with Backed-Up Values at MAX and MIN Levels

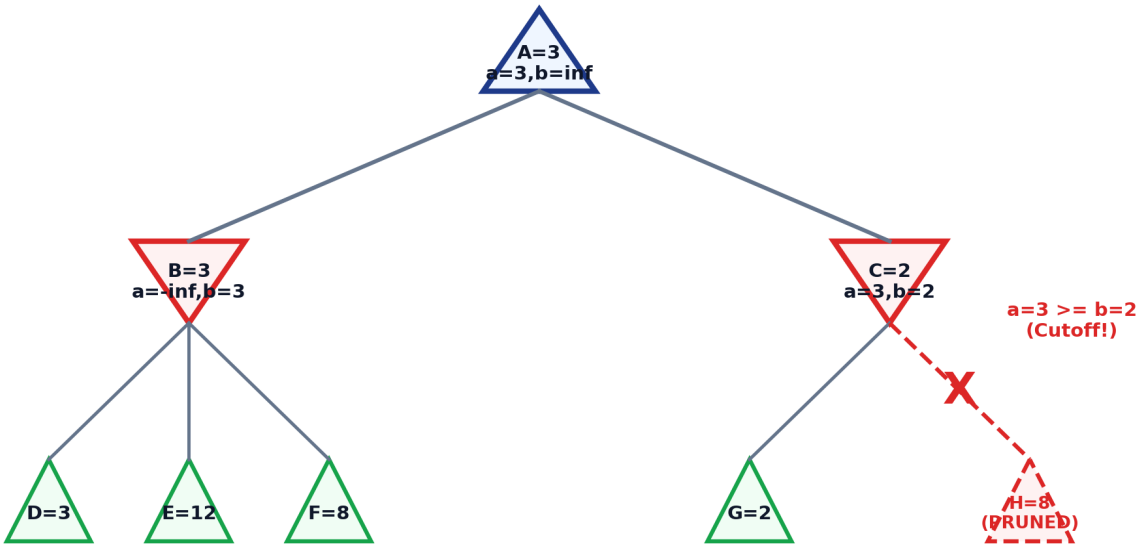


Figure 8: Alpha-Beta Pruning Execution Tree showing (a, b) Updates and Pruned Branch

Solved Numerical 5.1 (Actual PYQ — Alpha-Beta Pruning Execution) [10 Marks]

Source: Compiled NMIMS B.Tech CE Final Examination, Batch 2024-25, Q3.a [10 Marks]

Problem Statement: Trace Alpha-Beta Pruning on the game tree in Figure 8. Root **A** is MAX. Show (α, β) updates at every node, pruned subtrees, cutoff justifications, backed-up values, and optimal move.

ALPHA-BETA PRUNING EXECUTION LOG & CUTOFF PROOF

Step-by-Step Traversal & (α, β) Execution Log:
1. **Node A (MAX Root):** Initialized with $\alpha = -\infty$, $\beta = +\infty$.
2. **Node B (MIN Left Child):** Inherits $\alpha = -\infty$, $\beta = +\infty$.
- Evaluates Leaf D = 3 $\Rightarrow$ $\beta = \min(+\infty, 3) = 3$.
- Evaluates Leaf E = 12 $\Rightarrow$ $\beta = \min(3, 12) = 3$.
- Evaluates Leaf F = 8 $\Rightarrow$ $\beta = \min(3, 8) = 3$.
- Returns backed-up value **v(B) = 3**.
3. **Root A Update:** $\alpha = \max(-\infty, 3) =$ **3**. (MAX knows it can guarantee at least 3).
4. **Node C (MIN Right Child):** Inherits $\alpha = 3$, $\beta = +\infty$.
- Evaluates Leaf G = 2 $\Rightarrow$ $\beta = \min(+\infty, 2) =$ **2**.
- **Check Cutoff Condition:** $\alpha = 3 \geq \beta = 2 \Rightarrow$ **ALPHA CUTOFF TRIPPED AT NODE C!**
- **PRUNED BRANCH:** Leaf **H (val=8)** is skipped without evaluation.
- Returns backed-up value **v(C) = 2**.
5. **Root A Decision:** v(A) = max(v(B)=3, v(C)=2) = **3**.
• **Final Results:** Root Value **v(A) = 3** | **Optimal Move: Move to B** | **Pruned Leaves: Leaf H**.

6. RECEPTACLE OF COMMON EXAM MISTAKES & TRAPS

Exam Area / Topic	Common Student Failure Mode / Trap	Correct Method for 100% Full Marks
BFS Goal Testing	Applying goal test when popping node from queue.	Apply goal test when *generating* child node to save 1 full level expansion.
UCS Goal Testing	Applying goal test when generating child node.	Apply goal test when *popping node for expansion*; otherwise cheaper path missed!
8-Puzzle Blank Space	Including blank space as a misplaced tile in *h(n)*.	Exclusively count *numbered tiles 1 to 8*. Blank space is NOT a tile!
Alpha-Beta Cutoff	Confusing α (lower bound) and β (upper bound).	MAX updates α; MIN updates β. Cutoff trips as soon as $\alpha \geq \beta$.

7. FINAL NUMERICAL READINESS CHECKLIST & SYLLABUS AUDIT

100% UNIT 2 SYLLABUS READINESS VERIFICATION

• [x] **Problem Formulation:** 5-component framework (s0, Actions, Result, GoalTest, PathCost) memorized.
• [x] **Uninformed Search Traces:** BFS, DFS, UCS, DLS, IDS executed with OPEN/CLOSED lists.
• [x] **Complexities Derived:** BFS $O(b^{d+1})$, DFS $O(b^{m})$, UCS $O(b^{1+\blacksquare\ C^*/\varepsilon\ \blacksquare})$.
• [x] **Informed Search Traces:** GBFS $f(n)=h(n)$ and A* $f(n)=g(n)+h(n)$ solved step-by-step.
• [x] **8-Puzzle Solved:** Misplaced tiles *h1* and Manhattan distance *h2* grid transitions verified.
• [x] **Admissibility Proof:** Verified $0 \leq h(n) \leq h^*(n)$ for all nodes.
• [x] **Local Search Pathologies:** Local Maxima, Plateaux, and Ridges identified with remedies.
• [x] **Genetic Algorithm Worked:** Selection probability $p_i = f_i/\Sigma f$, crossover, and mutation calculated.
• [x] **Adversarial Game Trees:** Minimax utility backing-up and Alpha-Beta pruning ($\alpha \geq \beta$) cutoffs executed.
• [x] **Syllabus Audit:** 100% Unit 2 coverage verified against official course policy *Artificial Intelligence 2026-27.pdf*.